

**Decision Tree Regression is used to solve regression problems.**

**A regression problem** means we want to **predict a numerical value**.

| #  | Assessment | Project | Result |
|----|------------|---------|--------|
| 1  | Good       | Yes     | 95     |
| 2  | Average    | No      | 70     |
| 3  | Good       | Yes     | 75     |
| 4  | Poor       | No      | 45     |
| 5  | Good       | Yes     | 98     |
| 6  | Average    | Yes     | 80     |
| 7  | Good       | No      | 75     |
| 8  | Poor       | Yes     | 65     |
| 9  | Average    | No      | 58     |
| 10 | Good       | Yes     | 89     |

In Decision Tree Regression (DTR), we compute the standard deviation reduction (SDR) for all features, and the feature with the highest value becomes the root node, followed by the rest.

## **Decision Tree Regression — Steps**

**Identify the features** → Assessment, Assignment, Project.

**Identify the target** → Result.

**Calculate the standard deviation** of the Result data.

**Now we need to calculate the SDR of all features**

**Take 1<sup>st</sup> feature**

1. **Try different splits** for each feature.
2. **Calculate the standard deviation** for each resulting group.
3. **Calculate SDR** for each split.

**Repeat the Same for all feature.**

**Compare the SDR values.**

**Select the split with the highest SDR** as the root node.

# Calculate SD of Result

| #  | Assessment | Project | Result |
|----|------------|---------|--------|
| 1  | Good       | Yes     | 95     |
| 2  | Average    | No      | 70     |
| 3  | Good       | Yes     | 75     |
| 4  | Poor       | No      | 45     |
| 5  | Good       | Yes     | 98     |
| 6  | Average    | Yes     | 80     |
| 7  | Good       | No      | 75     |
| 8  | Poor       | Yes     | 65     |
| 9  | Average    | No      | 58     |
| 10 | Good       | Yes     | 89     |

$$\begin{aligned} \bar{x} &= \frac{95 + 70 + 75 + 45 + 98 + 80 + 75 + 65 + 58 + 89}{10} \\ &= \frac{750}{10} \\ \bar{x} &= 75 \end{aligned}$$

$$SD = \sqrt{\frac{(95 - 75)^2 + (70 - 75)^2 + (75 - 75)^2 + (45 - 75)^2 + (98 - 75)^2 + (80 - 75)^2 + (75 - 75)^2 + (65 - 75)^2 + (58 - 75)^2 + (89 - 75)^2}{10}}$$

Dividing by N

$$\frac{2464}{10} = 246.4$$

Taking the Square Root

$$\sqrt{246.4} \approx 15.70$$

## Calculate SD of Assessment

| #  | Assessment | Project | Result |
|----|------------|---------|--------|
| 1  | Good       | Yes     | 95     |
| 3  | Good       | Yes     | 75     |
| 5  | Good       | Yes     | 98     |
| 7  | Good       | No      | 75     |
| 10 | Good       | Yes     | 89     |

### A. Assessment = Good

Results:

95, 75, 98, 75, 89

### Average

$$Average = \frac{95 + 75 + 98 + 75 + 89}{5}$$

$$Average = 86.4$$

### Standard Deviation

$$SD_{Good} = \sqrt{\frac{(95 - 86.4)^2 + (75 - 86.4)^2 + (98 - 86.4)^2 + (75 - 86.4)^2 + (89 - 86.4)^2}{5}}$$

$$= \sqrt{\frac{475.2}{5}}$$

$$SD_{Good} = 9.75$$

## Calculate SD of Average

| # | Assessment | Project | Result |
|---|------------|---------|--------|
| 2 | Average    | No      | 70     |
| 6 | Average    | Yes     | 80     |
| 9 | Average    | No      | 58     |

### B. Assessment = Average

Results:

70, 80, 58

Average

$$Average = \frac{70 + 80 + 58}{3}$$

$$Average = 69.33$$

Standard Deviation

$$SD_{Average} = \sqrt{\frac{(70 - 69.33)^2 + (80 - 69.33)^2 + (58 - 69.33)^2}{3}}$$

$$SD_{Average} = 8.99$$

## Calculate SD of Poor

| # | Assessment | Project | Result |
|---|------------|---------|--------|
| 4 | Poor       | No      | 45     |
| 8 | Poor       | Yes     | 65     |

### C. Assessment = Poor

Results:

45, 65

Average

$$Average = \frac{45 + 65}{2} = 55$$

Standard Deviation

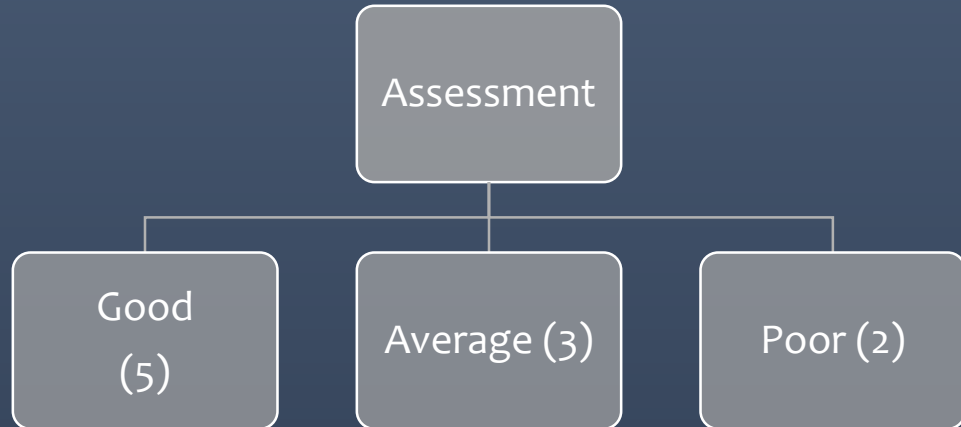
$$SD_{Poor} = \sqrt{\frac{(45 - 55)^2 + (65 - 55)^2}{2}}$$

$$= \sqrt{\frac{200}{2}}$$

$$SD_{Poor} = 10$$

## Calculate SDR of Feature one (Assessment)

| #  | Assessment | Result |
|----|------------|--------|
| 6  | Average    | 80     |
| 2  | Average    | 70     |
| 9  | Average    | 58     |
| 1  | Good       | 95     |
| 3  | Good       | 75     |
| 5  | Good       | 98     |
| 10 | Good       | 89     |
| 7  | Good       | 75     |
| 8  | Poor       | 65     |
| 4  | Poor       | 45     |



$$\text{Weighted SD} = \sum_{i=1}^k \frac{N_i}{N} \cdot SD_i$$

Where:

- $N_i$  is the number of instances in child subset  $i$
- $N$  is the total number of instances in the parent dataset ( $N = \sum N_i$ )
- $SD_i$  is the standard deviation of child subset  $i$

### Weighted Standard Deviation — Assessment

$$\begin{aligned} \text{Weighted SD} &= \frac{5}{10}(9.75) + \frac{3}{10}(8.99) + \frac{2}{10}(10) \\ &= 4.875 + 2.697 + 2 \end{aligned}$$

$$\boxed{\text{Weighted SD} = 9.57}$$

### Standard Deviation Reduction

$$SDR = 15.70 - 9.57$$

$$\boxed{SDR_{\text{Assessment}} = 6.12}$$

## 4. Project

Calculate the SD of Yes

| #  | Assessment | Project | Result |
|----|------------|---------|--------|
| 1  | Good       | Yes     | 95     |
| 3  | Good       | Yes     | 75     |
| 5  | Good       | Yes     | 98     |
| 6  | Average    | Yes     | 80     |
| 8  | Poor       | Yes     | 65     |
| 10 | Good       | Yes     | 89     |

### A. Project = Yes

Results:

95, 75, 98, 80, 65, 89

Average

$$Average = \frac{95 + 75 + 98 + 80 + 65 + 89}{6}$$

$$Average = 83.67$$

Standard Deviation

$$SD_{Yes} = \sqrt{\frac{(95 - 83.67)^2 + (75 - 83.67)^2 + (98 - 83.67)^2 + (80 - 83.67)^2 + (65 - 83.67)^2 + (89 - 83.67)^2}{6}}$$

$$SD_{Yes} = 11.54$$

## 4. Project

Calculate the SD of No

| # | Assessment | Project | Result |
|---|------------|---------|--------|
| 2 | Average    | No      | 70     |
| 4 | Poor       | No      | 45     |
| 7 | Good       | No      | 75     |
| 9 | Average    | No      | 58     |

### B. Project = No

Results:

70, 45, 75, 58

Average

$$Average = \frac{70 + 45 + 75 + 58}{4}$$

$$Average = 62$$

Standard Deviation

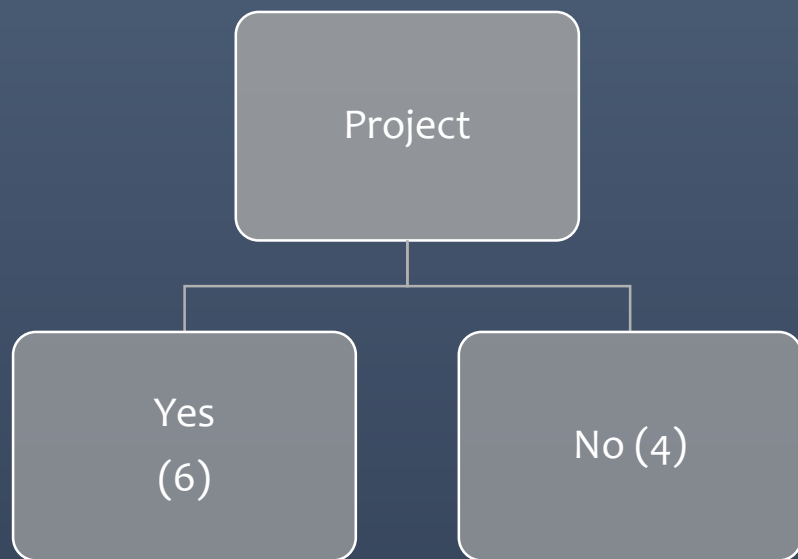
$$SD_{No} = \sqrt{\frac{(70 - 62)^2 + (45 - 62)^2 + (75 - 62)^2 + (58 - 62)^2}{4}}$$

$$= \sqrt{\frac{538}{4}}$$

$$SD_{No} = 11.60$$

## Calculate the SDR feature-2 Project

| Project | Result |
|---------|--------|
| Yes     | 80     |
| Yes     | 95     |
| Yes     | 75     |
| Yes     | 98     |
| Yes     | 89     |
| Yes     | 65     |
| No      | 70     |
| No      | 58     |
| No      | 75     |
| No      | 45     |



### Weighted Standard Deviation — Project

$$\begin{aligned} \text{Weighted } SD &= \frac{6}{10}(11.54) + \frac{4}{10}(11.60) \\ &= 6.924 + 4.64 \end{aligned}$$

$$\boxed{\text{Weighted } SD = 11.56}$$

### Standard Deviation Reduction

$$SDR = 15.70 - 11.56$$

$$\boxed{SDR_{\text{Project}} = 4.13}$$

| Attribute         | Weighted SD | SD Reduction |
|-------------------|-------------|--------------|
| <b>Assessment</b> | <b>9.57</b> | <b>6.12</b>  |
| Project           | 11.56       | 4.13         |

### Best Attribute

The attribute with the **highest Standard Deviation Reduction** is:

Assessment = 6.12

Therefore, for the **Decision Tree Regressor**, the first/root split should be:

**Assessment**

| #  | Assessment | Project | Result |
|----|------------|---------|--------|
| 6  | Average    | Yes     | 80     |
| 2  | Average    | No      | 70     |
| 9  | Average    | No      | 58     |
| 1  | Good       | Yes     | 95     |
| 3  | Good       | Yes     | 75     |
| 5  | Good       | Yes     | 98     |
| 10 | Good       | Yes     | 89     |
| 7  | Good       | No      | 75     |
| 8  | Poor       | Yes     | 65     |
| 4  | Poor       | No      | 45     |

